HACKATHON PITCH PROPOSAL • TRACK 3

VOTERVOICE NAIJA

Inclusive, Multilingual & Offline-Resilient Civic Technology for All Nigerians

Bridging the Electoral Divide with Conversational Speech AI, Zero-Data USSD/IVR, and Vote Protection Protections

No Internet Needed | Hausa / Yoruba / Igbo / Pidgin | Works on 2G Yam Phones | PWD Assistive Tech | Vote

1. Executive Summary & Core Vision

In Nigeria, genuine democracy requires universal access and ballot integrity. Today, over **40 million rural and semi-urban citizens** are systematically excluded from modern civic tech because existing platforms demand high-speed internet, smartphones, and English literacy. **VoterVoice Naija** solves this last-mile crisis with a **dual-channel civic platform** that bridges high-end generative AI with zero-internet feature phones ("Yam phones") over toll-free GSM voice (IVR) and USSD in **Hausa, Yoruba, Igbo, and Nigerian Pidgin**, ensuring every citizen can participate, understand accreditation rules, and legally protect their vote from invalidation.

2. The Problem Statement: The 4 Pillars of Civic Exclusion

Pillar A: The Hardware & Internet Divide

• Over 40% of adult Nigerians in rural zones own basic 2G feature phones without data plans.
• Traditional web portals (IREV, voter verification) require broadband and browser capability.
• Result: Tens of millions cannot verify polling units or confirm accreditation rules.

Pillar B: The Linguistic & Dialectal Chasm

• Nigeria has 500+ indigenous languages, yet over 90% of civic apps are published solely in English.
• Legalistic electoral terms confuse non-English speakers across rural communities.
• Grassroots voters become vulnerable to viral hoaxes, election falsehoods, and manipulation.

Pillar C: Text Non-Literacy & Oral Culture

• Millions of eligible voters are functionally illiterate in text, but highly articulate orally.
• Information transfer in rural Nigeria has always centered on oral discussions and radio.
• Text-heavy SMS and portals trigger intimidation, confusion, and polling-day disenfranchisement.

Pillar D: Systemic Exclusion of PWDs

• 25M+ Nigerians live with disabilities; millions face physical and informational barriers.
• Extreme lack of Nigerian Sign Language (NSL) resources for Deaf and hard-of-hearing voters.
• Visually impaired voters lack audio-guided ballot navigation, compromising their secret ballot.

Target Demographics & Quantified Human Impact

• **Rural Agricultural & Riverine Communities:** Farmers, traders, and artisans in Kebbi, Ebonyi, Ondo, Borno, and Bayelsa using 2G button phones.
• **Non-Literate First-Time Voters:** Young rural citizens navigating PVC collection, accreditation, and BVAS procedures without intimidation.
• **Voters with Disabilities (PWDs):** 25M+ citizens gaining independent civic agency through audio navigation and Nigerian Sign Language (NSL).
• **Grassroots Civil Society & Community Leaders:** Village heads and faith leaders equipped with verified, audio-broadcastable civic bulletins.

3. The Proposed Solution: Dual-Access Architecture

VoterVoice Naija operates on a **Zero-Barriers Design Doctrine**. Whether a voter owns a ₦4,000 feature phone in a remote village or an iPhone in Lagos, they receive identical, verified electoral information in their mother tongue through channels tailored to their reality.

Pillar 1: Zero-Internet IVR & USSD Telephony Gateway

- **Toll-Free Interactive Voice Response (IVR):** Voters dial a toll-free number (0800-VOTE-NG). They speak naturally in Hausa, Yoruba, Igbo, or Pidgin, and hear a clear spoken response.
- **USSD Fast-Track (*384*24#):** Interactive session menus for rapid polling unit search and election day alerts with zero data charge and zero internet.
- **DTMF Keypad Fallback:** Non-literate users can press 1 for Hausa, 2 for Yoruba, 3 for Igbo, 4 for Pidgin, following simple audio prompts.

Pillar 2: Multilingual Speech AI & Knowledge Grounding Core

- **Accented Automatic Speech Recognition (ASR):** Speech-to-text fine-tuned on African and Nigerian language corpora (Masakhane / MMS).
- **INEC-Grounded RAG Pipeline:** An authoritative vector repository indexed on the Electoral Act 2022, INEC Regulations, and official circulars.
- **Strict Anti-Hallucination & Non-Partisan Guardrails:** System strictly refuses political predictions, voter persuasion, or unverified claims.

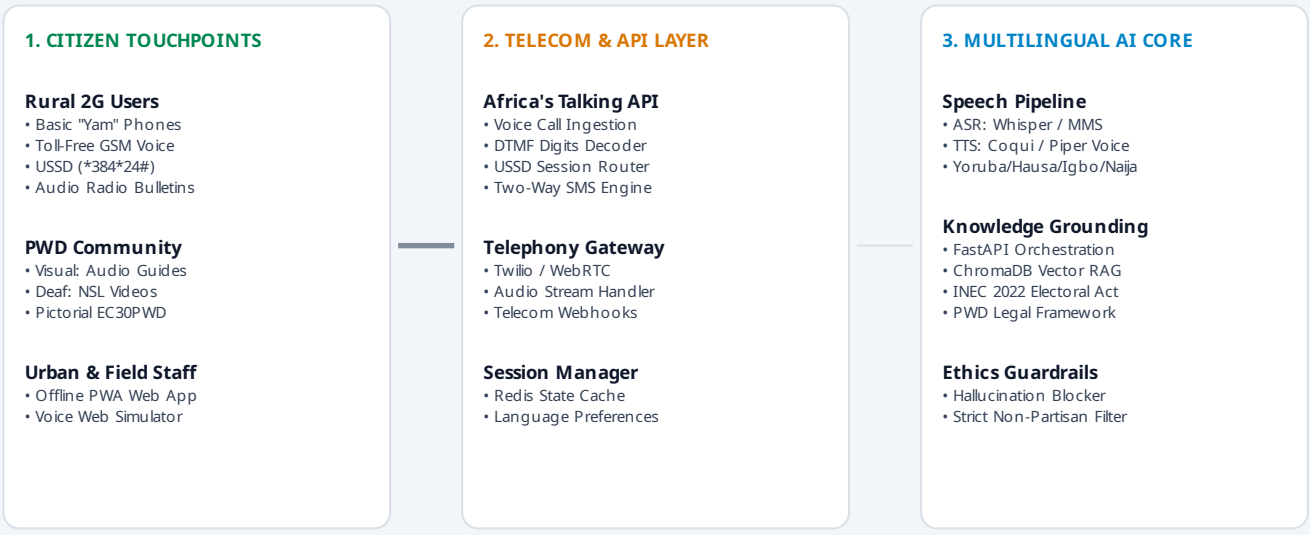
Pillar 3: Comprehensive PWD Assistive Suite

- **Nigerian Sign Language (NSL) Video Library:** High-definition animated video explainers interpreting voting procedures for the Deaf.
- **Digital Form EC30PWD:** Interactive pictorial guides visually illustrating accreditation, voting secrecy, and priority queuing.
- **Tactile & Audio Ballot Practice Simulator:** Accessible web/audio simulator enabling visually impaired voters to rehearse ballot layout.

Pillar 4: Offline-First Field Companion (Progressive Web App)

- **100% Offline Caching:** Local SQLite/IndexedDB caches all educational audio and guides for grassroots civic mobilizers in off-grid zones.
- **Store-and-Forward Telemetry:** Collects accessibility feedback and crowd queries offline; automatically syncs when connectivity is detected.
- **Low-Power Solar Hub Compatibility:** Packageable into an offline micro-server (Raspberry Pi) for distribution at village centers.

System Architecture: End-to-End Last-Mile Information Pipeline



4. Citizen Participation & Vote Protection Blueprint

In Nigerian elections, participating is only half the battle; **ensuring every ballot legally counts** is the true democratic imperative. VoterVoice Naija empowers rural, illiterate, and PWD voters across the **5 critical stages of voting integrity**, directly addressing rejected ballots, premature queue closures, and collation manipulation.

Stage 1: Pre-Voting Verification (Eliminating Wrong-Polling-Unit Disenfranchisement)

- **Participation Action:** Citizen dials toll-free 0800-VOTE-NG or USSD *384*24# in Hausa, Yoruba, Igbo, or Pidgin.
- **How Vote Counts:** Confirms exact Polling Unit (PU) code and location before election day. Eliminates arriving at the wrong PU on voting day, which causes non-accreditation and automatic loss of vote.

Stage 2: Ballot Literacy & Valid Thumbprinting (Eliminating Voided / Rejected Ballots)

- **Participation Action:** Audio/pictorial guide explains ballot marking techniques in indigenous languages.
- **How Vote Counts:** Directly combats the #1 cause of lost rural votes: smudged ink into adjacent boxes. Teaches the **'Inward Vertical Fold'** rule so wet indelible ink never transfers across to an opposing candidate's box, preventing ballot disqualification.

Stage 3: BVAS Accreditation Defense (Preventing Section 51 Over-Voting Cancellation)

- **Participation Action:** Citizens understand statutory BVAS rights under Section 47 of the Electoral Act 2022.
- **How Vote Counts:** Protects the polling unit against automatic cancellation. If ballots exceed BVAS-accredited voters, Section 51 mandates zeroing the unit. Educates voters to demand facial capture if fingerprints fail and guarantees accreditation occurs before receiving a ballot.

Stage 4: Queue Cut-Off Protection (Defending the 2:30 PM Statutory Right - Section 56)

- **Participation Action:** Real-time audio advice on voting hours and queue legal rights via feature phone.
- **How Vote Counts:** Election officials sometimes attempt to close queues prematurely. The system arms citizens with legal proof: **anyone in the queue by 2:30 PM has an inviolable right to be accredited and vote**, stopping unlawful disenfranchisement.

Stage 5: Open Count Witnessing & Form EC8A Auditing (Section 60 Collation Defense)

- **Participation Action:** Voters are encouraged to remain peacefully at the PU to witness the open counting and score announcements.
- **How Vote Counts:** Presiding Officers must openly count votes, paste Form EC8A, and upload to IREV. Citizens report announced tallies via toll-free voice note, creating a tamper-evident grassroots audit trail that prevents manipulation during transit to collation centers.

Real-World Walkthroughs: How Musa, Amina & Emeka Vote and Protect Their Ballots

Mallam Musa (Rural Sokoto)

54yo Farmer • 2G Yam Phone • Hausa Only

- **Dials *384*24#:** Confirms exact Polling Unit code in Gwadabawa without traveling.
- **Listens to Audio Guide:** Hears Hausa instructions on inward vertical ballot folding.
- **At the Booth:** Demands BVAS facial scan when dry farm fingers fail; marks ballot cleanly; vote is validly counted.

Amina (Deaf Voter in Oyo)

21yo Student • Budget Android • NSL User

- **Watches NSL Videos:** Offline PWA explains entire voting day sequence in Nigerian Sign Language.
- **Navigates Booth:** Recognizes official gestures; matches ballot colors (Green for Senate, Red for Reps).
- **Ballot Secrecy:** Votes 100% independently without relying on biased third-party escorts.

Emeka (Enugu Mobilizer)

38yo Community Leader • 2G Phone

- **Defends Queue:** Calls 0800-VOTE-NG at 2:15 PM; cites Section 56 to prevent premature queue shutdown.
- **Witnesses Count:** Observes open counting under Section 60; verifies Form EC8A pasted on wall.
- **Voice Audit:** Sends spoken tally report to citizen monitoring dashboard to deter result swapping.

The PWD Vote Secrecy & Integrity Guarantee

- **Visually Impaired Citizens:** Pre-election audio rehearsal allows blind voters to memorize tactile ballot positions, eliminating reliance on biased third-party escorts and preserving a secret ballot.
- **Deaf & Low-Literacy Citizens:** NSL video guides and Form EC30PWD pictorial instructions confirm ballot box identification, ensuring ballots go into the correct designated box.

5. Technical Stack & Human-Centric Design System

A. Production-Grade Technical Stack Specifications

LAYER / DOMAIN	TECHNOLOGY CHOICES & JUSTIFICATION
Telephony & Voice Gateway	**Africa's Talking Voice & USSD API** (Direct integration with MTN, Airtel, Glo, 9mobile in Nigeria) + **Twilio / WebRTC** for international judge demos.
Speech Recognition (ASR)	**OpenAI Whisper-large-v3** fine-tuned on Nigerian accents + **Meta MMS (Massively Multilingual Speech)** supporting Hausa, Yoruba, and Igbo phonemes.
Retrieval-Augmented AI (RAG)	**FastAPI + LangChain / LlamaIndex** connected to **ChromaDB / pgvector**, embedding verified INEC Electoral Acts, Manuals, and Official Gazettes.
Speech Synthesis (TTS)	**Piper TTS & Coqui TTS** (ultra-fast, CPU-runnable local speech models) + **Google Cloud Text-to-Speech** for localized Nigerian English and Yoruba accents.
Backend Services & Cache	**Python 3.11+ / FastAPI** for high-throughput asynchronous request handling, with **Redis** for persistent IVR call state and **PostgreSQL** for audit logs.
Frontend & Simulator	**Next.js 14 (React), Tailwind CSS, Lucide Icons**, with **Workbox PWA** for offline caching and an interactive on-screen feature phone dialer.
Accessibility & PWD Suite	**HTML5 Video with NSL (Nigerian Sign Language)**, High-Contrast UI Mode (WCAG 2.1 AA certified), Web Speech API for browser voice input.
Deployment & Edge Support	**Docker & Docker Compose** for local containerization; **Render / AWS ECS** for cloud hosting; optional **Raspberry Pi 4** offline kiosk package.

B. Human-Centric Design Philosophy & Color System

Civic technology must feel warm, human, and encouraging—never sterile, dark, or bureaucratic. Our visual system is rooted in **Nigerian civic authenticity, warm human earth tones, high-contrast readability**, and compliance with **WCAG 2.1 AA standards**.

Civic Emerald
#008751
Brand & Trust
Evokes Nigerian sovereignty, democracy, and verified integrity.

Warm Terracotta
#E07A5F
Human Warmth
Approachability, communal connection, and inclusive hospitality.

Golden Ochre
#D97706
Clarity & Vitality
Action items, important deadlines, and engaging highlights.

Friendly Sky
#0284C7
Assistance & Tech
Informational badges, audio playback indicators, and links.

Warm Alabaster
#F4F5F0
Paper Canvas
Soft natural background; prevents eye fatigue in glaring sunlight.

Core UI/UX Accessibility Standards (Human-Centered Design)

- **Contrast Ratio:** Minimum 7.1:1 contrast on all text elements, guaranteeing clarity on low-brightness phone screens.
- **Touch Target Ergonomics:** 48px minimum button size with audio-touch feedback for tremor/elderly users.
- **Cognitive Load Reduction:** Maximum 3 choices per audio menu; spoken prompts repeat twice with slow cadence.
- **Zero Text-Dependency:** All primary interactions supported via speech or universally recognized non-text iconography.

6. Team Roles (Team of 4) & Execution Roadmap

To build and submit a winning, fully functional proof-of-concept within the hackathon timeline, tasks are cleanly distributed across 4 specialized roles without overlapping bottlenecks.

Member 1: Lead AI & Speech Systems Engineer

Core Owner: Conversational AI Pipeline & Multilingual Models

- Builds the FastAPI ASR/TTS pipeline for Hausa, Yoruba, Igbo, and Pidgin.
- Sets up ChromaDB vector store and indexes the Electoral Act 2022 & INEC Guidelines.
- Implements strict anti-hallucination prompting and non-partisan guardrails.
- Delivers clean REST endpoints: /api/transcribe, /api/query-electoral-rag, /api/synthesize.

Member 2: Telephony, USSD & Backend Engineer

Core Owner: Real-Time Telecom Infrastructure & Session Management

- Sets up Africa's Talking / Twilio voice sandbox and provisions Nigerian virtual numbers.
- Builds the USSD menu callback handler (*384*24#) in Python/FastAPI.
- Implements Redis-based session caching for stateful multi-turn IVR phone calls.
- Connects inbound audio streams from telecoms to Member 1's AI speech pipeline.

Member 3: Accessibility UI/UX & Web Simulator Engineer

Core Owner: Frontend Demo, Accessible Design & PWA Architecture

- Develops the Next.js 14 web simulator featuring an on-screen interactive 2G feature phone.
- Implements the PWD accessibility suite (Nigerian Sign Language video player & EC30PWD).
- Configures Workbox Progressive Web App (PWA) for 100% offline caching and field testing.
- Ensures strict compliance with WCAG 2.1 AA contrast and screen-reader accessibility.

Member 4: Civic Research, Content Translation & Demo Lead

Core Owner: Ground Truth Data, Translation Auditing & Submission Package

- Curates and verifies official INEC documents, FAQs, and polling unit directory datasets.
- Records native speech sample prompts in Hausa, Yoruba, Igbo, and Nigerian Pidgin.
- Validates system answers against Nigerian electoral law to eliminate hallucinations.
- Plans, scripts, and produces the mandatory 2-minute submission video and pitch deck.

Sprint Execution Roadmap (Towards Grand Finale)

Sprint 1: Architecture & Data Grounding

- Initialize Git monorepo with FastAPI backend and Next.js frontend.
- Ingest and embed INEC Electoral Act 2022 and voter FAQ documents into ChromaDB vector database.
- Provision Africa's Talking developer account and configure local ngrok webhook tunneling.

Sprint 2: Multilingual Speech & Telecom Integration

- Wire up speech-to-text (Whisper/MMS) and text-to-speech for Hausa, Yoruba, Igbo, and Pidgin.
- Implement the USSD *384*24# interactive menu and IVR voice call callback routes in FastAPI.
- Test live GSM voice calls on Africa's Talking simulator.

Sprint 3: PWD Modules, Offline PWA & Web Simulator

- Build the on-screen feature phone dialer in Next.js to simulate 2G calls directly in the browser.
- Integrate Nigerian Sign Language (NSL) video assets and digital Form EC30PWD pictorial guides.
- Enable offline service workers and cache validation for field testing.

Sprint 4: Rigorous Testing, Video Production & Submission

- Run non-partisan ethical audit and edge-case testing with native speakers.
- Record and edit the **2-minute demonstration video** showcasing both feature-phone voice and web UI.
- Complete final deployment on cloud instance and submit project on ai4elections.risenetworks.org.

7. Developer Setup Manual: What & What to Install

Follow this step-by-step installation manual to get the complete VoterVoice Naija stack running on local development machines.

Step 1: System Prerequisites (All Team Members)

Ensure the following core runtimes and development tools are installed on your operating system:

- **Python 3.11+:** Core backend & AI runtime (verify with **`python3 --version`**)
- **Node.js 18+ & npm:** Modern JavaScript frontend runtime (verify with **`node -v`**)
- **Git:** Version control for team collaboration (verify with **`git --version`**)
- **ngrok:** Local secure tunneling to expose webhooks to Africa's Talking telephony (**`ngrok http 8000`**)

Step 2: Backend & AI Engine Installation (Members 1 & 2)

Clone the project repository and set up a dedicated Python virtual environment:

```
git clone https://github.com/your-team/votervoice-naija.git && cd votervoice-naija/backend
python3 -m venv venv && source venv/bin/activate  # On Windows: venv\Scripts\activate
```

Install all required Python packages with pip:

```
pip install fastapi uvicorn pydantic python-dotenv langchain langchain-community \
            chromadb openai tiktoken africas-talking redis requests gTTS python-multipart
```

Configure your local **`.env`** file:

```
AT_USERNAME=sandbox
AT_API_KEY=your_africas_talking_sandbox_api_key
OPENAI_API_KEY=your_api_key  # For Whisper ASR & RAG embeddings
REDIS_URL=redis://localhost:6379/0
```

Step 3: Frontend & Web Simulator Installation (Member 3)

Navigate into the frontend directory and initialize the Next.js application:

```
cd ../frontend
npm install next react react-dom lucide-react clsx tailwindcss postcss autoprefixer
npm install @radix-ui/react-dialog @radix-ui/react-tabs @radix-ui/react-slider
npm install next-pwa workbox-window  # For offline Progressive Web App caching
```

Configure **`frontend/.env.local`**:

```
NEXT_PUBLIC_API_URL=http://localhost:8000
NEXT_PUBLIC_ENABLE_NSL_VIDEO=true
```

Step 4: Africa's Talking Sandbox Setup (Member 2)

1. Create a free developer account at **`account.africastalking.com`** and switch to the **Sandbox app**.
2. Navigate to **Voice > Phone Numbers** and generate a free simulated Nigerian phone number.
3. Navigate to **USSD > Service Codes** and create channel code **`*384*24#`**.
4. Start your local ngrok tunnel: **`ngrok http 8000`**.
5. Set your USSD Callback URL in Africa's Talking to: **`https://your-ngrok-url.ngrok-free.app/api/ussd/callback`**.
6. Set your Voice Callback URL to: **`https://your-ngrok-url.ngrok-free.app/api/voice/callback`**.

Step 5: Launching & Testing the Prototype

1. Start the FastAPI AI Backend:

```
cd backend && source venv/bin/activate && uvicorn main:app --reload --port 8000
```

2. Start the Next.js Frontend & Phone Simulator:

```
cd frontend && npm run dev  # Opens http://localhost:3000
```

3. Test 2G Feature Phone Simulator: Open **`http://localhost:3000/simulator`** to dial *384*24# or simulate an audio voice call.

4. Test Inbound GSM Call: Launch Africa's Talking WebRTC Simulator to place a real call to your sandbox number.